CHANDU PRASAD BHAIRAPU



Agentic AI Developer & GenAI Engineer

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ABOUT ME

AI developer with an MSc in Artificial Intelligence and 3 years of professional experience in Generative AI, agentic AI and machine learning. I build software with embedded AI agents and multi-agent systems using LangGraph and MCP across the full development lifecycle, from solution architecture and testing to deployment on AWS and Azure with CI/CD in GitLab. I develop AI-assisted with Claude Code and my own custom skills.

EXPERIENCE

Associate Generative AI Developer – Agentic Indexing Pipeline Engineer (Energy Sector)

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction pipeline end-to-end (data ingestion, chunking, prompt engineering, and indexing with agentic, deterministic, and monolithic strategies), reporting directly to the CTO and implementing requirements with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph with millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, GraphRAG, RAG, Neo4j, MongoDB, Postgres, FastAPI, React, Azure, Kubernetes, Docker**

GenAI Developer (AI Consulting Intern) – Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed an agentic AI application with LangGraph that orchestrates requests across vector database queries, general inference, and summarization, and deployed it on AWS.
- Built the full stack: React frontend, Python backend with FastAPI, API testing with Postman.
- Containerized the entire stack with Docker and deployed it to AWS using CDK; version-controlled in GitLab with a CI/CD pipeline.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman**

GenAI Developer (Master's Thesis) – Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG generation and retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, PyTorch, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search, Docker, Streamlit**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, engineering lag features and rolling statistical windows from sensor data.
- Benchmarked models against empirical curve-fitting to identify deviating batches earlier; deployed a Streamlit dashboard letting chemists simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer – Electric Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Contract Risk Agents | Strands Agents | A2A | MCP | RBAC

- Multi-agent contract review system: a clause-extraction agent and a compliance-check agent (Strands Agents, Amazon Bedrock) communicate over the A2A protocol; contract data and compliance rules are exposed through an MCP server.
- Role-based access control (RBAC) on agent tools, OpenTelemetry tracing of every agent and tool call, a JavaScript web UI, and a GitLab CI/CD pipeline with automated tests.

TECHNICAL SKILLSET

- **Agentic AI & GenAI:** LangGraph, LangChain, MCP, multi-agent systems, RAG, GraphRAG, RAG evaluation (RAGAS), prompt engineering, Hugging Face Transformers
- **Programming:** Python, SQL; JavaScript for React frontends
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time-series analysis
- **Cloud & DevOps:** AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3, VPC), Azure, Docker, Kubernetes, GitLab CI/CD, Git
- **AI Coding Assistants:** Claude Code (custom skills, agentic workflows)
- **Databases:** Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search), SPARQL

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science — Aug 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

CONFERENCE PUBLICATIONS

[1] Navigating the Future: An approach of Autonomous Indoor Vehicles

CERTIFICATIONS

- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

German (B1) | English (Business Fluent) | Hindi (Fluent) | Telugu (Native)